claude-windows / 45ea3a91-654d-4972-9cd3-65f35db54caf

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\45ea3a91-654d-4972-9cd3-65f35db54caf\subagents\workflows\wf_2a64bc88-2a8\agent-aef6a2fa8789081cf.jsonl`
- SHA-256: `fc9bc51d551a569df9d979d9c551cb414f35d17448b84d29fce793d45938ccd5`
- Source modified: `2026-06-10T05:43:43+00:00`
- Imported at: `2026-07-05T16:48:27+00:00`
- project: `wf_2a64bc88-2a8`
- session_id: `45ea3a91-654d-4972-9cd3-65f35db54caf`

Transcript

1. user (2026-06-10T05:41:39.830Z)

CONTEXT ? two sibling repos on a Windows machine:
- INDEXER: C:/Users/avidu/Projects/badminton-highlight-indexer ? local AI badminton(->racket-sports) highlight indexer + rally detector. FastAPI + SQLite + ffmpeg + GPU CV via WSL; Gemini = paid cloud AI. docs/ tree is rich. Current branch docs/next-steps-multisport (master = main branch).
- COLLECTOR: C:/Users/avidu/Projects/sports-data-collector ? golden-label acquisition/curation/ingestion CLI (system-of-record candidate for training corpus).
Project state (June 2026): scaffolding complete; owned-model roadmap agreed (docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md) but NO platform/owned-model implementation started; next committed platform slice = async job model. Licensing guardrail: MIT/Apache-2.0/BSD only for code+data+weights; paid LLMs are inference-only, never training-data sources.
ENVIRONMENT RULES: this dev environment has NO GPU/torch/cv2 ? do NOT attempt to run the video pipeline or import torch/cv2 modules. Test suites are offline/fully-mocked and SAFE to run. You are READ-ONLY: never edit/create/delete repo files; running tests and git read commands is allowed.
QUALITY BAR: cite file paths (and line numbers where possible) for every finding. Report what IS in the code/docs, not what you assume. Be skeptical and concrete. Your final structured output is consumed programmatically by an orchestrator ? return raw data, not prose for a human.
ROLE: adversarial verifier. An auditor ("docs-strategy") claimed the finding below. Try to REFUTE it against the actual files. Default to isReal=false if the evidence does not hold up, the behavior is documented/deliberate (e.g. listed in docs/CODE_MAP.md gotchas or DEFERRED_HARDENING_PLAN.md as a known deferral ? unless the auditor adds genuinely NEW consequence), or the severity is materially overstated.
FINDING: {"title":"Four docs give four different answers to 'what is next': async job model vs Phase-0 M0.x vs Packaging-first vs ROADMAP Phase 4","severity":"high","area":"sequencing / roadmap","file":"C:/Users/avidu/Projects/badminton-highlight-indexer/CLAUDE.md:20","evidence":"CLAUDE.md:20: 'Next slice = async job model (DEFERRED_HARDENING_PLAN.md #16), single-tenant' (checkpoint README:30 agrees). DEFERRED_HARDENING_PLAN.md:12-14: 'Recommended sequence: #7 Packaging (lowest risk...) ? #13 Train/eval guardrails ? #16 Async jobs (largest)' ? async LAST within its own doc. COMMERCIALIZATION.md §6 priority order (lines 167-180) ranks 'SaaS / multi-user / async foundations (C12)' #7 of 8, 'lower until monetization is firm'. ROADMAP.md:107 status table marks 'Phase 4 Offline learned segmenter' as '? Next'. NEXT_STEPS.md:28 puts '[owner greenlight] Phase-0 (M0.x)' as priority #1, with only a footer caveat (line 46) 'Committed next PLATFORM slice = async job model'. The plan's footer (lines 164-165) tries to reconcile: 'the committed next platform slice remains the async job model...; only the M0.1?M0.4 foundation is safe to start' ? but CLAUDE.md simultaneously says 'Do not start implementation without explicit owner approval' with no mention of M0.x at all.","why_it_matters":"A reader landing on any single doc would start different work (packaging vs async vs TAL harness vs offline segmenter). The decision authority (owner greenlight per milestone) is stated only in the plan/NEXT_STEPS, not in CLAUDE.md; and CLAUDE.md's 'next slice = async' contradicts COMMERCIALIZATION's own ranking of async as near-last priority."}
Steps: open the cited file(s) and any counter-evidence files; verify quotes/line claims; judge

severity honestly. Cheap greps allowed; do NOT run the pipeline. Return isReal, confidence, and a 1-3 sentence verdict (with corrected detail if partially right).

2. user (2026-06-10T05:41:47.491Z)

```
# CLAUDE.md ? project memory / agent handoff

Local, AI-powered **badminton (? racket-sports) highlight indexer + rally detector**:
FastAPI + SQLite + ffmpeg + GPU CV/AI, with paid **Gemini** as the cloud AI. Sibling
repos: `sports-data-collector`, a VLC-based `rally-annotator`.

## Read these first
- **`docs/README.md`** ? doc index. **`docs/CODE_MAP.md`** ? code navigation + data
  contracts (the cold-start map; read before touching code).
- **Strategy (post-scaffolding):** `docs/COMMERCIALIZATION.md`, `docs/PMF_MARKET_RESEARCH.md`,
  `docs/PLATFORM_ARCHITECTURE.md`, `docs/DEFERRED_HARDENING_PLAN.md`. Also `docs/ROADMAP.md`
  (offline-segmenter narrative) and `docs/BACKLOG.md`.
- **History:** `docs/archives/` ? ADRs (`decisions/DECISIONS.md`), research, and
  **`checkpoints/`** (latest: `2026-06-strategy-foundation/`). Archive policy: only move
  **terminal-state** (DONE/REJECTED) work; live docs stay at `docs/` top level.

## Current state (June 2026)
- **Scaffolding complete**; the **post-scaffolding strategy foundation** is laid and merged
  (#52/#53/#55/#57/#58). **No platform/owned-model implementation has started.**
- **Next slice = async job model (`DEFERRED_HARDENING_PLAN.md` #16), single-tenant.**
  **Do not start implementation without explicit owner approval.**

## Architecture in one breath
- **Pluggable registries** (ABC + `Registry` singleton): segmenters / providers / validators
  / sports / annotations ? see `backend/pipeline/segmenters/base.py`. *Future:*
  `StorageBackend` + `ComputeBackend` registries (PLATFORM §3).
- `video_id` = MD5 of the file ? `backend/utils/hashing.py::compute_video_id` (one helper).
- Typed config = `backend/config/` (Pydantic). DB schema evolves via the `PRAGMA
  user_version` migration ladder in `backend/infrastructure/database.py`.

## Committed guardrails (non-negotiable)
- **Paid LLMs (Gemini/OpenAI/Anthropic) = inference / serving / fallback ONLY ? never a
  training data source** (terms/copyright). Training teachers/bases = **open Apache-2.0 only**.
- **Owned-model base:** Apache-2.0, video-capable (Qwen-VL / Gemma 4); **reject Gemma 3**
  (viral license). **Exclude minors** from the training pool; per-user training data needs a
  separate explicit opt-in.
- **v1 = rally + highlight + history + correction loop**; **gait/biometrics strictly future**
  (GDPR Art. 9).
- **Generic data schema**; `coach ? athlete` is an **optional overlay**, not baked in.

## Workflow conventions
- **Branch off `master`; ONE PR per work item; NO PR stacking.** Don't open a PR unless asked.
- Tests: `python run_all_tests.py` (offline, fully mocked). Python 3.8+; `ffmpeg` required.
- Never push to the default branch directly; descriptive commit messages.
- ?? This dev container lacks GPU / torch / cv2 / numpy ? the **full pipeline can't run here**.
  Verify pure-logic paths and hand machine-side checks (real video, GPU/WSL suite) to the owner.

```

3. user (2026-06-10T05:41:47.845Z)

Next steps (live) ? owned-model + multisport

Updated: 2026-06-09 (end of the quality-iteration + multisport-recon session). **Authoritative roadmap:**
`docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md`. **Latest blow-by-blow:** memory `current-state.md` / `session-handoff.md`.
This file = the prioritized "what to do next," refreshed each session.

The recommendation (one paragraph)
Build the **quality-first extraction harness** before anything else: a learned **TAL head** (ActionFormer/ASFormer, MIT)
over trajectory features is the robust path ? it **decouples from the multi-court "background games" confound** that the
unsupervised heuristic (and audio, and spatial court-gating) can't escape (all three tested + failed; it's a *temporal*
rally-vs-non-rally problem). **Framework = ms-swift, base = Qwen3-VL** (Gemma-4 AMBER/bench) come **later** for the
conversational/grounding upgrade ? start with the head. **On-device is deferred** (export-clean ops now so it stays
reachable). Everything stays **MIT/Apache-2.0/BSD** (code + data + weights). Bottleneck = **data + per-sport trajectory
detectors**, not model architecture.

Latest findings (2026-06-09)
- **6-match badminton LOVO:** learned prototype **0.583** vs heuristic **0.480** (learned leads since n=3, AUC 0.83?0.92,
  decoupled from multi-court). The "contrast" metric (in-rally÷dead-time visibility) predicts heuristic F1 monotonically.
- **?? Multisport (Roora pickleball + TT):** labels **ingest cleanly** (`curate convert-cvat` ? 6 rallies/sport, sane).
  WASB-transfer SURPRISE (directional, n=6 each ? TINY): the badminton shuttle detector segments **TABLE TENNIS at
  F1 0.909** out-of-the-box (!) but **pickleball only 0.308**. ? **The per-sport-detector blocker is SPORT-DEPENDENT:** TT may
  be servable with WASB short-term; pickleball needs a proper (MIT/Apache) ball detector. Confirm/deny with more videos.
- **Owner field notes (2026-06-09):** camera height/distance **varied on purpose** (great generalization data); **multi-court
  adjacent-court SOUND dominates the signal** ? empirically reconfirms audio-foreground isolation won't work + the
  robustness-first learned-model path. Multi-court is the realistic (academy) environment, so this *is* the target distribution.

Prioritized next steps
1. **[owner greenlight] Phase-0 (M0.x), $0/local:**
   - **M0.4 ? Gen-0 TAL harness** (first-class): productionize `rally_seq_proto.py` ? one-command train?eval?**ship-gate** over
     ActionFormer/ASFormer, reusing `metrics.match_intervals`. **Define the ship-gate target** (explicit F1@tIoU; floor = beat
     heuristic LOVO 0.480) ? "very high P/R" must be operationalized, not just "beat the floor."
   - **M0.1 ? corpus spine + the per-video event-index contract** (the conversational seam: extract ? event store ? text-to-SQL).
   - **M0.3 ? compliance:** purge Gemini-derived training labels (`training.label_candidates`); WASB **C3** own-weights/clean-swap (gates ship).
2. **Multisport thread:**
   - Ingest the Roora pickleball/TT rally CSVs into the corpus (collector `import-local`).
   - **Get more pickleball + TT videos** ? re-run the WASB-transfer eval to confirm/deny TT?0.91, pickleball?0.31 (n=6 is too few).
   - Once each sport has ?3 matches, run a **merged prototype LOVO across badminton + TT (+ pickleball)** ? the "how would a
     merged model do" experiment. Identify a clean MIT/Apache ball detector for sports

```
where WASB doesn't transfer (pickleball).
3. **Housekeeping:** **rename repo off "badminton"** (multisport); confirm collector = corpus/event-store system-of-record.

## Pointers / artifacts
- Roadmap: `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` · Study: `docs/archives/research/OWNED_MODEL_TRAINING_STUDY.md`
  (2026-06-09 banner) · Quality: `docs/archives/quality-iterations/2026-06-multivideo-lovo/`.
- Eval corpus: `output/human_lovo_manifest.json` (badminton n=6). Roora CSVs: `%TEMP%\roora\{pickleball,tabletennis}_rallies.csv`;
  WASB-transfer trajectories: `%TEMP%\{pickleball,tabletennis}_traj.csv`. Code: `backend/eval/{eval_partial_labels,rally_seq_proto,calibrate_local}.py`.
- ? No owned-model implementation until owner greenlights a milestone. Committed next PLATFORM slice = async job model (single-tenant).
```

4. user (2026-06-10T05:41:55.524Z)

```
# Deferred Hardening Plan ? Train/Eval Guardrails, Async Jobs, Packaging

**Status:** ? Design for review ? **no code yet**. Owner-requested follow-up to the
"config & identity consistency" tech-debt pass. Each item below becomes its own PR
only after this doc is approved.

**Why these three:** they were cataloged as out-of-scope during the consistency
pass because each is larger/design-heavy or needs the owner's machine. They are the
remaining high-value hygiene items not already done (config/structure/Result/logging/
reports) or in flight (detector abstraction, PR #49).

**Recommended sequence:** #7 Packaging (lowest risk, unblocks clean installs/CI) ?
#13 Train/eval guardrails (correctness of the data flywheel) ? #16 Async jobs
(largest, product-facing).

---

## 1. Packaging modernization (debt #7)

### Problem
- No `pyproject.toml` on `master`. (PR #49's description lists packaging as "done",
  but it isn't present ? reconcile with the owner.)
- The root `setup.py` is a **diagnostics/dependency-checker**, not a PEP 517 build
  script ? its name collides with setuptools semantics and will confuse `pip`/`build`.
- Deps live only in `requirements.txt`; `obsreport` is a commented soft dep
  (BACKLOG) and `supervision<0.30.0` is pinned (debt #14).

### Approach
- Add `pyproject.toml` (PEP 621): project metadata + runtime deps mirroring
  `requirements.txt`, with optional-dependency groups:
```

5. user (2026-06-10T05:41:56.469Z)

```
### PMF risks (from the readiness review)
- **Capture quality is the gate.** Accuracy is capture-conditional; broadcast/handheld
  footage is a failure mode. The wedge (static, tripod, full-court) is defensible but
  narrows the addressable market ? enforce/guide it (C11, `VIDEO_RECORDING_GUIDELINES`).
- **Value beyond raw segments is table stakes.** Time-saved editing alone is weak;
  per-rally tags, stats, and a correction loop are what make it sticky (C10).
- **Golden labels are the recurring blocker** across the offline segmenter, audio
  fusion, and measurable quality ? the highest-leverage investment (C8).

### Refined priority order (quality/data phase ? "focus on quality once new videos land")
Carried from the readiness review; commercialization items above are tracked in parallel.
```

```

1. **Golden-set expansion + data-flywheel UX** (highest with new labeled videos):
   ingest the new batch, expand calibration/CV/regression baselines, make label
   contribution + safe retraining easy and non-contaminating (eval/training separation).
2. **Hybrid unification + observability hooks** (deferred item 7, now higher): extract
   shared logic from `tracknet_hybrid`/`wasb_hybrid` before heavy AI timing/metrics land.
3. **Detector/windowing quality iteration** with new data: re-tune velocity/merge/
   min-duration + `rally_gate`; validate WASB on real video; trial audio fusion (ADR-007).
4. **Deeper value layer** (beyond minimal exports, C10): per-rally tagging/notes ?
   `candidate_segments`, per-player/match stats, clip gallery, human-correction loop.
5. **Input-quality enforcement & guidance polish** (C11): more guideline dimensions ?
   validators/report rules; "your video is good enough" UX.
6. **Multi-sport maturation + recording best practices.**
7. **SaaS / multi-user / async foundations** (C12) ? lower until monetization is firm.
8. **Polish & distribution** (C9): installer story, first-run path, codec/output, economics.

**Recommendation:** with the mid-week labeled-video batch, spend ~80% on #1 + #3
(flywheel + quality iteration on real data), then unification (#2) and deeper value (#4).

---

## 7. Open questions needing owner / counsel sign-off ??

1. **Ultralytics Enterprise price** ? only relevant if a permissive detector ever proves
```

6. user (2026-06-10T05:41:57.302Z)

```
- **Sport-generic.** The label ? feature ? classifier machinery and the export
  bridge already span sports; adding a sport is data, not new architecture.
- **Commercially clean.** Only `is_commercial` data flows through the bridge, so the
  trained model and any derived product rest on a licensable foundation.

## Status at a glance

| Phase | What | Status |
|------|------|--------|
| 1 | Export bridge (`curate export` ? manifest + CSVs) | ? Done |
| 2 | 1080p acquisition (`fetch`, in-place, cookies) | ? Done (22/22) |
| 3 | Eval + tuning loop (`run_phase3_eval.py`) | ? In progress |
| 4 | Offline learned segmenter (WASB shuttle-motion substrate) | ? Next (substrate decided: WASB, ADR-004) |
| + | Audio hit-detection cue (`backend/pipeline/audio/`, ADR-007) | ? E1 probe run ? AMBER, gated on golden labels |

> **Cross-cutting:** golden-set ground truth (which powers Phases 3?4 and on-demand regression testing)
> is produced via a vendor process + pluggable annotation provider ? see
> [GOLDEN_SET_VENDORING.md](GOLDEN_SET_VENDORING.md) and **ADR-006**. As of 2026-06-02 the annotation
> guideline is authored (collector `docs/annotation/`) and **Roora** (Bengaluru) is selected for an
> initial **3-sport pilot** (badminton, pickleball, table tennis). Input-capture quality requirements
> that bound achievable precision/recall are tracked in [VIDEO_RECORDING_GUIDELINES.md](VIDEO_RECORDING_GUIDELINES.md).
>
> **Observability:** every processed video now emits an always-on per-video diagnostics
> report (spec-driven warnings/errors with FAQ refs, soft-dependency on `obsreport`), so
> pipeline regressions and footguns surface without re-running eval ? see
```

7. user (2026-06-10T05:42:06.155Z)

150 | 1 | Greenlight Phase-0 milestone(s) ? rec: **M0.1 corpus+event-index**, then **M0.4 harness** | awaiting owner |
151 | 2 | WASB C3 ? first-class action item (own-weights / clean swap) | **owner: yes** ? |
152 | 3 | Ratify licensing **MIT / Apache-2.0 / BSD** (code+data+weights) | **owner: yes** ? |
153 | 4 | Collector = corpus + event-store system-of-record | **owner: confirm?** |
154 | 5 | Rename repo off "badminton" | **owner action** ? |
155 | 6 | Conversational-QA: build seam now, punt model | **rec accepted?** |
156
157 ## Open risks / unknowns (none block Phase-0 research)
158 Budget/cost to reach Gen-2 **unquantified** (largest risk) · ship-gate "very high P/R" target undefined · per-sport clean
159 trajectory detectors not yet identified · base-model pretraining provenance unauditable (risk-acceptance) · real VLM
160 VRAM/token cost · single-model-multisport-temporal generalization unproven · DPA + no-train (minors-excluded) for cloud
161 training · on-device 4B latency on 8 GB unmeasured.
162
163 ---
164 *Design-only and reversible. Nothing starts until you greenlight a milestone. Sequencing discipline holds: the committed
165 next platform slice remains the async job model (single-tenant); only the M0.1?M0.4 foundation is safe to start.*
166

8. user (2026-06-10T05:42:06.618Z)

1 # Roadmap ? from annotated data to a self-improving, offline rally segmenter
2
3 This document is the **narrative thread**: how the indexer evolves from an
4 AI-API-dependent prototype into a self-contained tool that learns rally detection
5 from its own commercially-licensed, annotated video corpus. It explains *how we
6 got here*, *why each step matters*, and *how it shapes the final tool*. Decisions
7 are recorded as ADRs in DECISIONS.md; the scoring mechanics live
8 in EVALUATION.md.
9
10 ## The thesis
11
12 The indexer detects badminton rallies in long recordings. Its strongest mode
13 (`yolo_hybrid`) is a two-stage pipeline: cheap **permissive person-detection
14 player-motion** filtering (Stage 1, local ? torchvision Faster R-CNN + ByteTrack
15 tracking) followed by **Gemini** semantic refinement (Stage 2, paid API). That
16 works, but it costs money, needs the network, and never improves on its own.
17
18 > *Naming note:* the `yolo_hybrid` registry key is retained for back-compat, but
19 > AGPL YOLOv8 was removed and replaced with a BSD/MIT-licensed torchvision +
20 > ByteTrack stack ? see **ADR-005**. "YOLO" in this doc means the local

9. user (2026-06-10T05:42:07.391Z)

1 # Checkpoint: 2026-06 Strategy Foundation (post-scaffolding)
2
3 **Date:** June 2026
4 **STATUS: COMPLETED**
5 **Trigger:** Owner merged the post-scaffolding strategy docs ? #52 (config/identity cleanup), #53 (deferred-hardening design), #55 (commercialization tracker), #57 (PMF market study), #58 (platform architecture).
6
7 ## Purpose
8 Captures the state once the **strategic / direction-setting foundation** for the
9 post-scaffolding phase was laid: the commercial dimension, market/PMF, the multi-user
10 platform architecture (pluggable storage/compute + the owned-model layer), and the
11 deferred-hardening designs. **Strategy only ? no implementation started.** Provides a
12 regression-detection baseline for priorities/decisions as quality + implementation work begins.

Key living documents at checkpoint time
- **COMMERCIALIZATION.md** ? commercial tracker (dashboard C1?C14, grouped Pending ? In-flight ? Completed with a P0/P1/P2 column).
- **PMF_MARKET_RESEARCH.md** ? cited market study; beachhead = **India badminton academies**.
- **PLATFORM_ARCHITECTURE.md** ? multi-user "assistant coach" platform: pluggable `StorageBackend`/`ComputeBackend` (BYO), generic schema (coach overlay optional), owned video-QA model layer (§5A), training-backend survey + nanochat-style harness, committed legal guardrail, per-user-adapter direction, v1-vs-vision scoping.
- **DEFERRED_HARDENING_PLAN.md** ? packaging, train/eval guardrails, async job model (#16) ? *pending approval*.
- **CODE_MAP.md / BACKLOG.md / ROADMAP.md** ? updated for the config/identity cleanup.

See git history around #52/#53/#55/#57/#58 for exact snapshots.

Committed decisions / non-negotiables recorded here
- **Paid frontier models (Gemini/OpenAI/Anthropic): inference / serving / fallback only ? never a training data source** (competing-model terms). Teachers/bases = open Apache-2.0 only.
- **Owned-model base = Apache-2.0-clean, video-capable** (Qwen2.5-VL-7B/32B, Qwen3-VL, Gemma 4); **reject Gemma 3** (viral license clause); **minors excluded** from the training pool; per-user training needs a separate opt-in.
- **v1 = rally detection + highlight compilation + per-user history + coach/user correction loop** ("assistant-coach lite"); **gait/biomechanics strictly future** (Art. 9).
- **Generic data schema** (User ? VideoCollection ? Video ? Highlight ? HighlightMetadata); coach ? athlete is an **optional overlay** (a market-fit hypothesis).

What comes next (per owner)
- First implementation slice = **async job model (DEFERRED #16)**, single-tenant ? **pending explicit go-ahead** (no implementation started without approval).
- Quality iterations with new labeled videos; golden-set expansion + the **golden QA eval set** (the prerequisite for the owned-model training harness).

When the first implementation slice or a quality milestone completes, add a new dated
checkpoint here (or under `quality-iterations/`).

10. user (2026-06-10T05:42:49.557Z)

8-- PMF_MARKET_RESEARCH.md ? Cited market study: sizing, customer segments, competitor pricing, and the recommended beachhead (feeds COMMERCIALIZATION §6).
9-- PLATFORM_ARCHITECTURE.md ? Strategy for the multi-user "assistant coach" platform: decoupled layers + pluggable storage/compute (BYO), multi-tenant data model, orchestration, and a phased roadmap.
10:- DEFERRED_HARDENING_PLAN.md ? Design (pending approval) for the larger deferred items: packaging, train/eval guardrails, and the async job model (#16, the first infra slice toward the platform).
11-- BACKLOG.md ? Deferred items with rationale and revisit triggers.
12-- ROADMAP.md ? High-level four-phase product direction.

11. user (2026-06-10T05:42:50.730Z)

37:| C12 | **SaaS foundations (async/auth/tenancy)** | **P1** | ? | Sync `/api/process` blocks on long videos; no auth/tenancy. | Multi-user/platform strategy in `PLATFORM_ARCHITECTURE.md` (pluggable storage/compute, tenancy, orchestration); async slice = `DEFERRED_HARDENING_PLAN.md` (#16). |
179:7. **SaaS / multi-user / async foundations** (C12) ? lower until monetization is firm.
221:- `DEFERRED_HARDENING_PLAN.md` ? packaging (C9), async (C12) designs (pending approval).

12. user (2026-06-10T05:42:57.388Z)

docs/

Live documentation for the badminton-highlight-indexer.

```
## For new contributors / ramp-up (read these first)
- [CODE_MAP.md](CODE_MAP.md) ? Complete navigation map of the codebase and data contracts.
- [COMMERCIALIZATION.md](COMMERCIALIZATION.md) ? Living tracker for the commercial dimension (licensing, codec/output economics, hosting, vendoring, packaging) + PMF risks and the quality/data-phase priorities. Supersedes the archived `COMMERCIAL_READINESS_REVIEW.md`.
- [PMF_MARKET_RESEARCH.md](PMF_MARKET_RESEARCH.md) ? Cited market study: sizing, customer segments, competitor pricing, and the recommended beachhead (feeds COMMERCIALIZATION §6).
- [PLATFORM_ARCHITECTURE.md](PLATFORM_ARCHITECTURE.md) ? Strategy for the multi-user "assistant coach" platform: decoupled layers + pluggable storage/compute (BYO), multi-tenant data model, orchestration, and a phased roadmap.
- [DEFERRED_HARDENING_PLAN.md](DEFERRED_HARDENING_PLAN.md) ? Design (pending approval) for the larger deferred items: packaging, train/eval guardrails, and the async job model (#16, the first infra slice toward the platform).
- [BACKLOG.md](BACKLOG.md) ? Deferred items with rationale and revisit triggers.
- [ROADMAP.md](ROADMAP.md) ? High-level four-phase product direction.
- [VIDEO_RECORDING_GUIDELINES.md](VIDEO_RECORDING_GUIDELINES.md) ? Capture best practices (referenced by reports).
- [HOW_RALLY_DETECTION_WORKS.md](HOW_RALLY_DETECTION_WORKS.md) ? Mental model of the two-layer detection system.
- [EVALUATION.md](EVALUATION.md) ? How we measure and improve quality.

## Practical / ongoing
- [TRACKNET_WSL_SETUP.md](TRACKNET_WSL_SETUP.md)
- Golden set docs (`GOLDEN_SET_*`)
- Evaluation harness and calibration drivers (in `backend/eval/`)

## Historical documents
Completed plans, old research, ADRs, and checkpoints live in [`archives/`](archives/).

See `archives/README.md` for the umbrella policy and `archives/past_projects/README.md` for classification of small tech debt / starter projects.
```

13. user (2026-06-10T05:43:39.181Z)

Structured output provided successfully

14. assistant (2026-06-10T05:43:43.675Z)

Verification complete.